Augmentation and leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable augmentation and source-lineage record.

Justify labels,audit both group/pixel overlap and compare declared training budgets.

Keep one habit: W1–W3/W5 → selected new grouped split → project provenance ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Return to the binary image task: class 1 means the left column sum is strictly larger than the right column sum; equal sums belong to class 0. Let d be left sum minus right sum.

  An **augmentation** is useful only under a stated task and transformation rule. More image files do not automatically mean more independent **evidence**.

- Swapping rows preserves each column sum, so it preserves this task label for every valid image. Swapping columns changes d to -d. For nonzero d the class changes; for d = 0 it stays class 0.

  Blindly copying the original label after every flip would create incorrect **training** targets.

- A transformation may be valid for one task and invalid for another. A crop can remove the **evidence** needed for a label.

  Choose allowed transformations using the task definition and **training evidence**, document the assumptions, and evaluate the choice on validation data.

Training a small image model

> **My remembered idea, example and correction**

Convolution as a local operation

> **My remembered idea, example and correction**

Debugging shapes and gradients

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Operation on rows [1, 0] and [0, 0]	Result rows	d	Class
Original	[1, 0] and [0, 0]	1	1
Swap top and bottom rows	[0, 0] and [1, 0]	1	1
Swap left and right columns	[0, 1] and [0, 0]	-1	0

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Task: class 1 iff the left-column sum is strictly greater than the right; ties are 0. Original [[1, 0], [0, 0]] has d=1 and class 1. Its row swap [[0, 0], [1, 0]] preserves the label; its column swap [[0, 1], [0, 0]] has d=-1 and class 0. Tie [[1, 0], [0, 1]] remains class 0 after column swapping, so always complementing a label is invalid. Transform functions assume valid 2×2 binary inputs and preserve originals.

  Week 23 E/F are row-swapped A/C arrays; training on those variants would duplicate validation pixels. Group by source before derivation; record parent, transformation, target and role. Separate P–T registry:training P/Q originals plus row swaps give 4 records from 2 groups; validation R original gives 1 record/group; final S/T originals give 2 records/groups. Total:7 retained, 3 derived evaluation rows excluded.

  The separate pictured A–D policy keeps two records per source and has only 1 final source. With one-example batches, 12 originals×3 records×5 epochs=180 updates, versus 12×5=60 for originals only. A baseline 15-epoch cycle matches 180 updates. Eight originals×3 records×4 epochs=96 updates, versus 8×4=32. No new training score is supplied.

- For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Task: class 1 iff the left-column sum is strictly greater than the right; ties are 0. Original [[1, 0], [0, 0]] has d=1 and class 1. Its row swap [[0, 0], [1, 0]] preserves the label; its column swap [[0, 1], [0, 0]] has d=-1 and class 0. Tie [[1, 0], [0, 1]] remains class 0 after column swapping, so always complementing a label is invalid. Transform functions assume valid 2×2 binary inputs and preserve originals.

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- For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Task: class 1 iff the left-column sum is strictly greater than the right; ties are 0. Original [[1, 0], [0, 0]] has d=1 and class 1. Its row swap [[0, 0], [1, 0]] preserves the label; its column swap [[0, 1], [0, 0]] has d=-1 and class 0. Tie [[1, 0], [0, 1]] remains class 0 after column swapping, so always complementing a label is invalid. Transform functions assume valid 2×2 binary inputs and preserve originals.

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- **Training** contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original.

  Identify the overlap and explain what must be rerun if this validation set already selected a model.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Task: class 1 iff the left-column sum is strictly greater than the right; ties are 0. Original [[1, 0], [0, 0]] has d=1 and class 1. Its row swap [[0, 0], [1, 0]] preserves the label; its column swap [[0, 1], [0, 0]] has d=-1 and class 0. Tie [[1, 0], [0, 1]] remains class 0 after column swapping, so always complementing a label is invalid. Transform functions assume valid 2×2 binary inputs and preserve originals.

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- Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches.

  Give both update counts and one way to match budgets without using final outcomes.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

  Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Design a grouped split independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Add a labelled **augmentation** and **leakage evidence** sheet to Image-pattern Network Investigation. Show how you can check whether a transformation preserves the task label.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Keep the equality boundary for **Augmentation** and **leakage**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Augmentation** and **leakage**. Proposed explanation: "Every image flip preserves the label". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

  Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Keep every augmented image with its original **source group**.
- Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/24

Augmentation and leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Track provenance

Each flipped copy retains its source ID and parent image reference.



2. Assign by source

A/B originals and flips give 4 training records; C gives 2 validation; D gives 2 final.



3. Check label meaning

Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

CHANGE ONE THING

Move D's flipped copy into training.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Keep every augmented image with its original **source group**.
- Which labelled step matches this detail? Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.
- 1. Assign by source
- 2. Track provenance
- 3. Check label meaning
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
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> ---
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> ---
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> ---

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check label meaning Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Changed-case question and optional hint

- Changed-case practice: Move D's flipped copy into **training**. Which conclusion is supported by the supplied facts?
- 1. Final-test group separation fails.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ____
>
> ____
>
> ____
>
> ____

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Final-test group separation fails. The transformed D image is still derived from the held-out source, despite a new filename.

Repair and retain the record

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

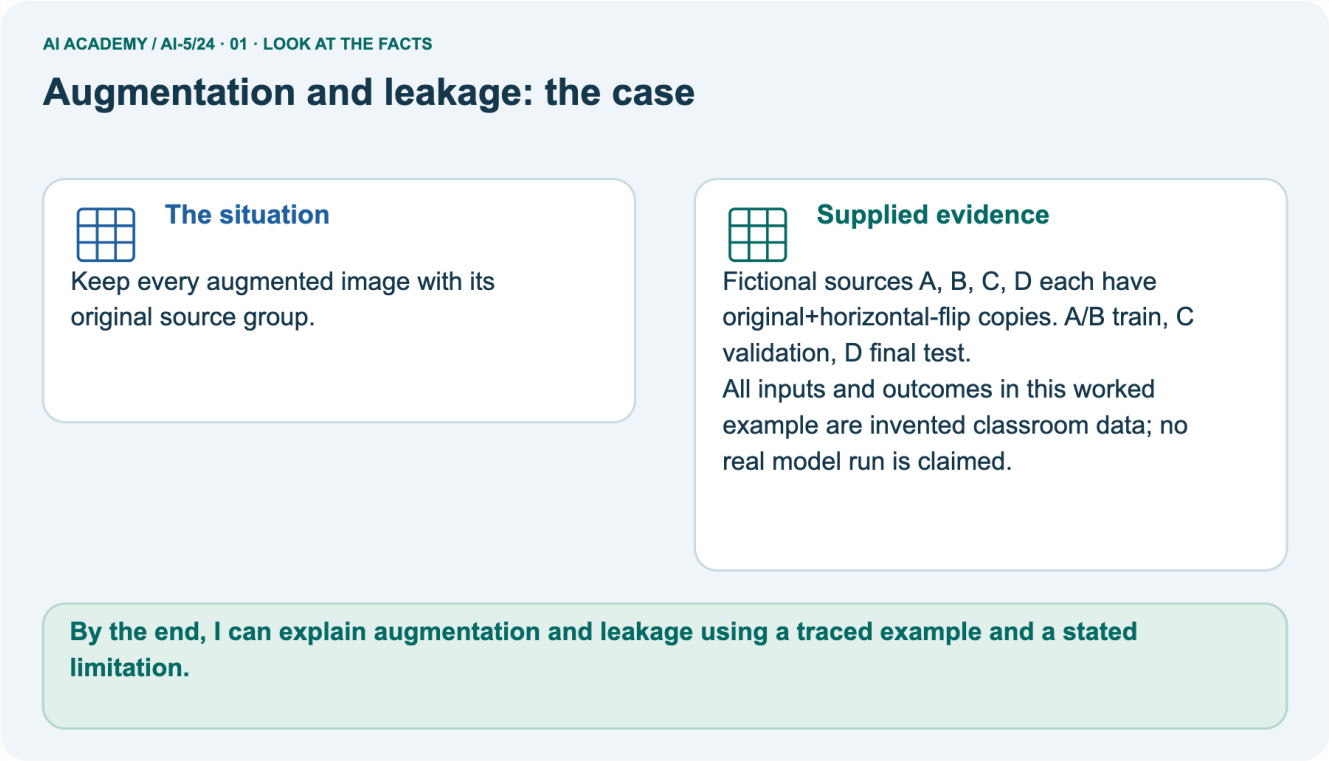
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Augmentation and leakage: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

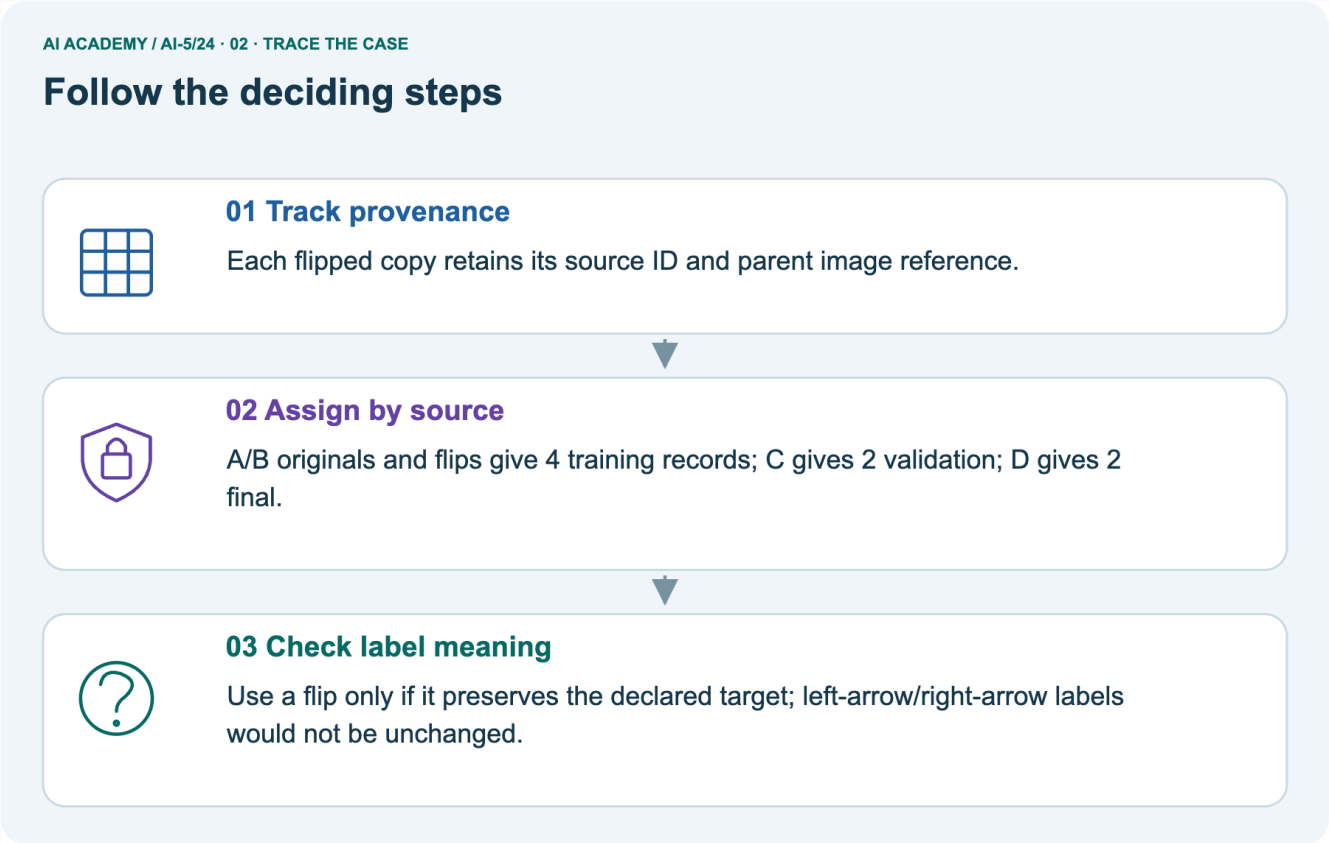
Keep every augmented image with its original **source group**. Supplied facts: Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Track provenance: Each flipped copy retains its source ID and parent image reference. → Assign by source: A/B originals and flips give 4 **training** records; C gives 2 validation; D gives 2 final. → Check label meaning: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Every variant stays with its source group

AI ACADEMY / AI-5/24 · 03 · INSPECT THE LESSON MODEL

Every variant stays with its source group

Source group	Assigned split	Derived records
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

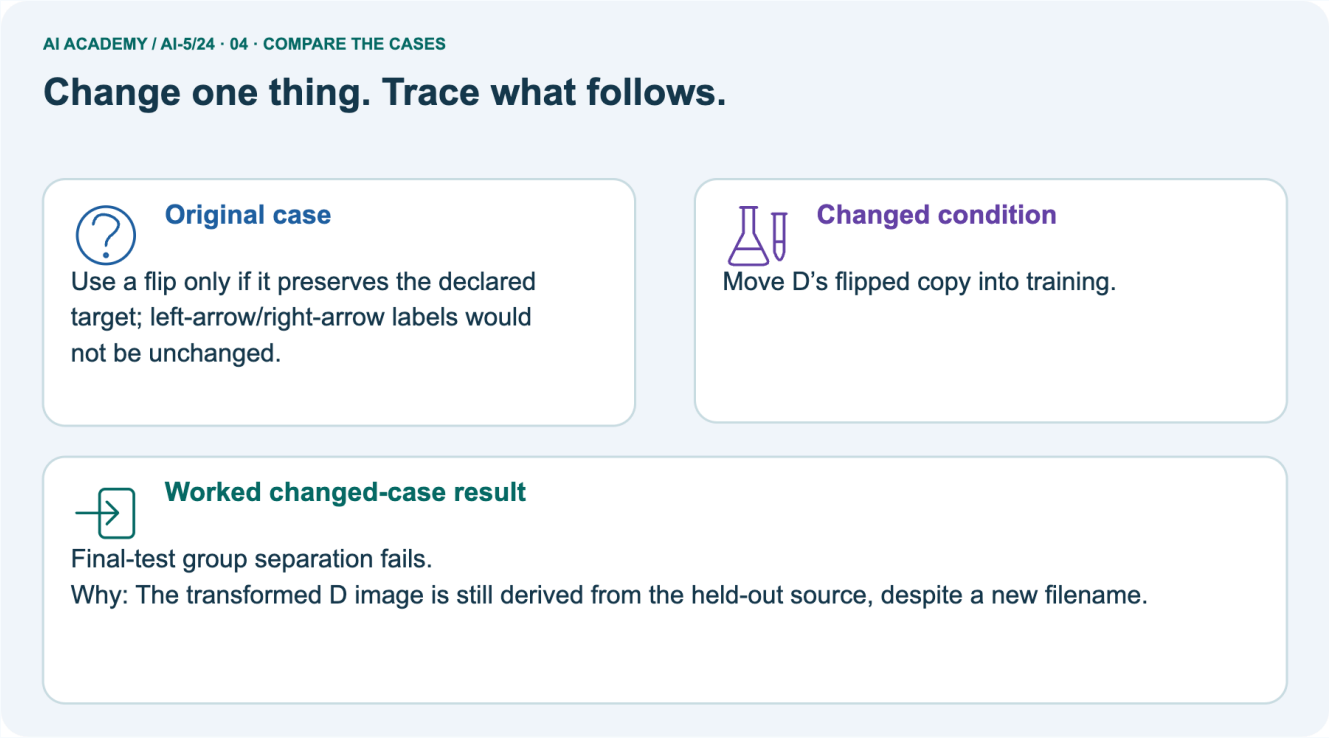
Follow the labelled picture with the matching explanation. Open the vector diagram

Every variant stays with its **source group**. A | **Training** | A original and A row swap. B | Validation | B original only. C | Final test | C original only

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

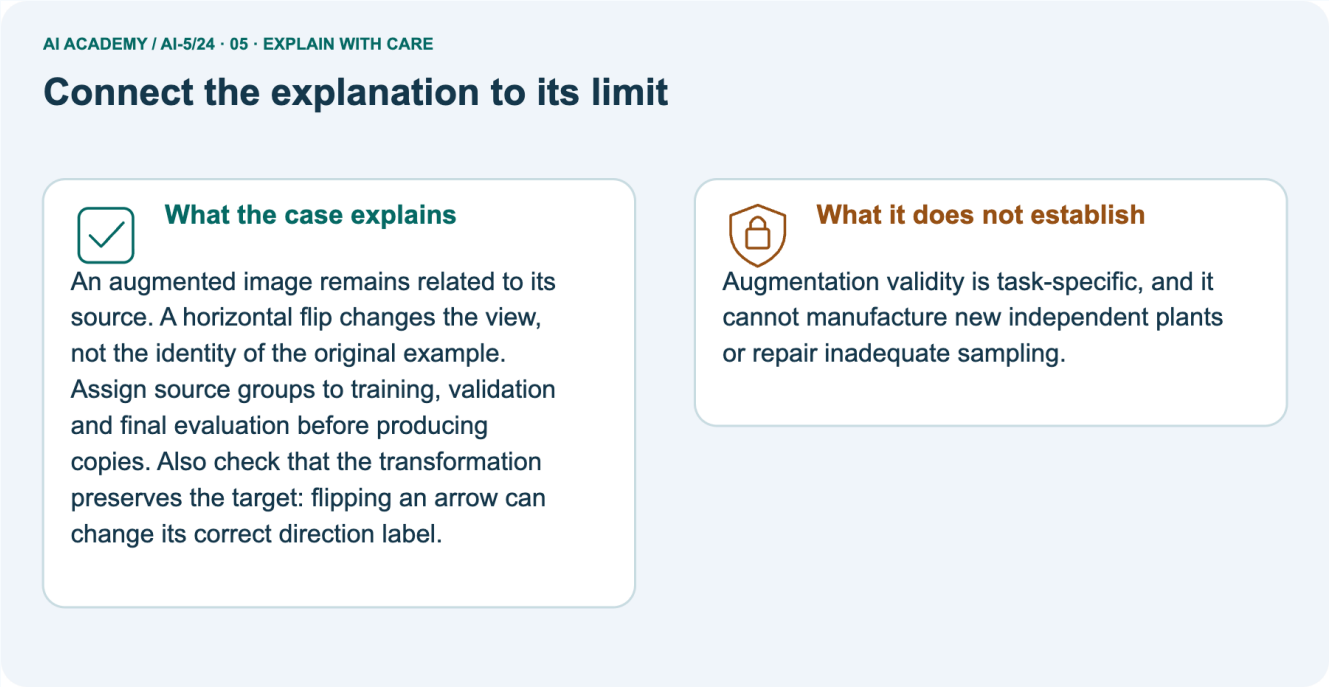
Original case: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged. Changed condition: Move D's flipped copy into **training**.

Expected result: Final-test group separation fails. Reason: The transformed D image is still derived from the held-out source, despite a new filename.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

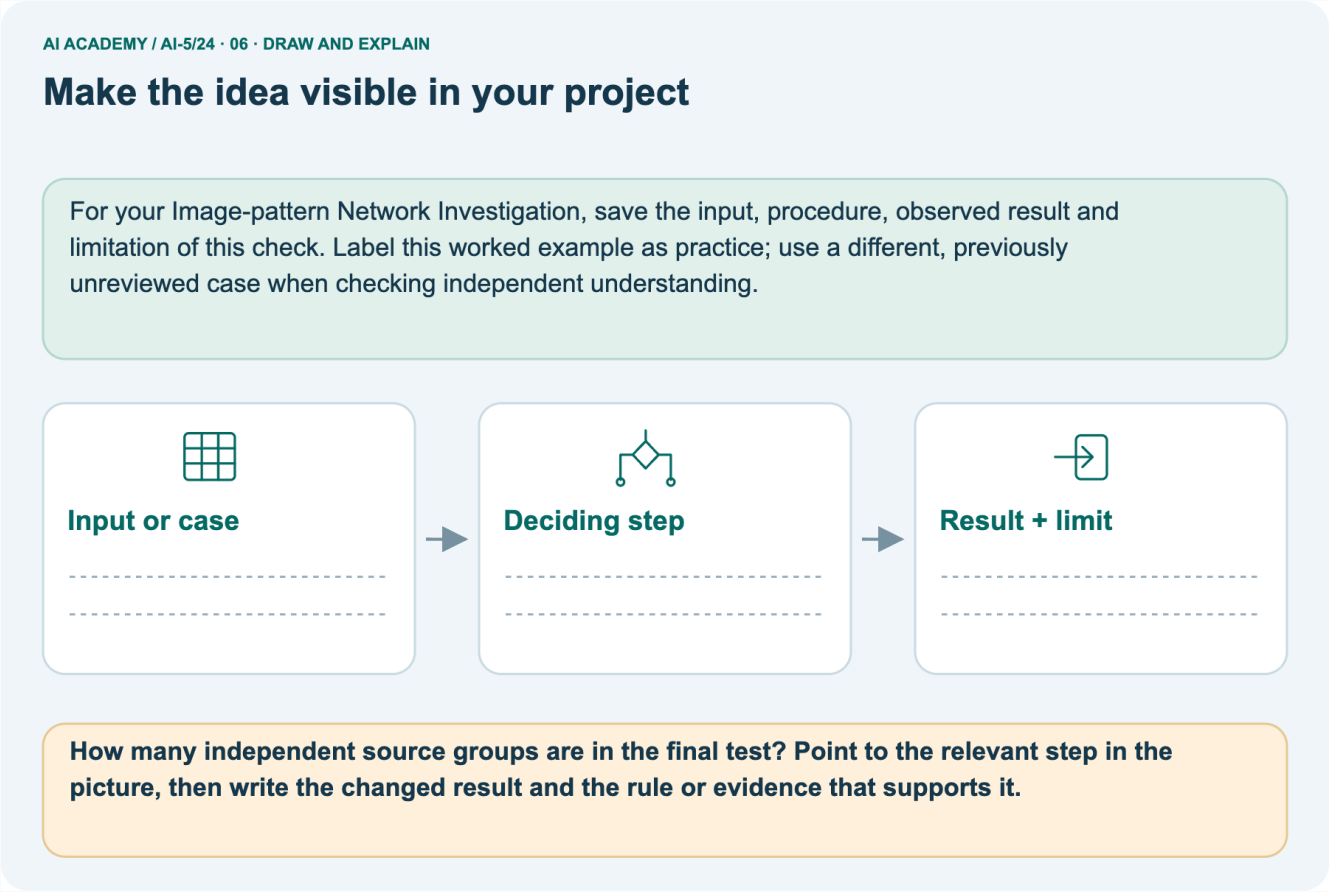
Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Limit: **Augmentation** validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Augmentation and leakage: the case

02. Follow the deciding steps

03. Every variant stays with its source group

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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